Abstract

We present RegionRAG, a retrieval-augmented generation system that grounds large language model answers in exact regions of source documents. Unlike conventional chunk-level RAG, RegionRAG retains page coordinates for every passage so that user interfaces can highlight the precise text supporting each claim.

In evaluation on 1,200 questions over technical reports, RegionRAG improved citation accuracy by 31 percent over a strong baseline and reduced unverifiable answers by half.

1. Introduction

Large language models frequently produce fluent but unsupported statements when answering questions about long documents. Retrieval-augmented generation mitigates this by conditioning the model on retrieved passages, but standard pipelines discard the spatial information needed to show users where evidence came from.

DocPilot is a voice-first application built on RegionRAG. Users upload documents, ask questions by voice or text, and receive answers linked to the exact supporting passages in the original file.

2. Method

Documents are parsed with a layout-aware extractor that records bounding boxes for each text region. Pages are split into overlapping chunks of roughly 220 words; chunks never span pages so each citation resolves to a single highlight rectangle.

Each chunk is embedded with a dense encoder and indexed with FAISS. At query time we fuse dense cosine similarity with BM25 lexical scores using reciprocal rank fusion, which recovers technical terms that dense encoders sometimes underweight.

The generator is instructed to answer only from retrieved evidence and to cite chunk identifiers. A server-side validator drops any cited identifier that was not actually retrieved, preventing the model from inventing page numbers or coordinates.

3. Experiments

We evaluate on three corpora: arXiv computer science papers, internal engineering reports, and product manuals. Metrics include answer correctness judged by experts, citation precision, end-to-end latency, and abstention quality on unanswerable questions.

RegionRAG achieves 88 percent citation precision versus 67 percent for chunk-only RAG. Median latency is 2.4 seconds for retrieval and generation combined, with retrieval accounting for 310 milliseconds.

4. Results

Ablations show that removing the lexical branch reduces citation precision by 9 points on manuals containing rare part numbers. Removing server-side citation validation increases hallucinated citations from 2 percent to 14 percent.

Users completed document question tasks 38 percent faster with source highlighting enabled, and reported higher trust when answers displayed the supporting excerpt next to the viewer.

5. Limitations

Our method depends on a usable text layer. Scanned documents require optical character recognition, and OCR errors propagate into chunks, embeddings, and highlighted regions. This build does not yet include an OCR fallback, so image-only PDFs are rejected during ingestion.

Bounding box quality varies with document structure. Tables, multi-column layouts, and math-heavy pages can produce fragmented reading order, which occasionally yields chunks that mix unrelated sentences and dilute retrieval quality.

The retrieval threshold is a single global constant. Documents written in very different styles may need per-document calibration, and we have not studied automatic threshold tuning.

Evaluation covers English documents only. Code-switched text and right-to-left scripts remain untested, and the speech interface inherits the recognition errors of the underlying browser engine.

6. Conclusion

Preserving the link between an answer and its original evidence is the core requirement for trustworthy document assistants. RegionRAG demonstrates that lightweight spatial metadata plus server-side citation validation is sufficient to make RAG answers verifiable.